Algorithms Students Must Implement

Dataset: Student Performance Factors

Students must use the provided dataset to implement all six machine learning algorithms. They must preprocess categorical features, handle missing values, and evaluate each model.

Task 1 — Linear Regression

Objective: Predict `Exam_Score`.

- Use `Hours_Studied`, `Attendance`, `Previous_Scores`, `Sleep_Hours`, and other relevant features.
- Convert categorical columns into numerical values using One-Hot Encoding.
- Split the data into training and testing sets.
- Train a Linear Regression model and make predictions.
- Evaluate using MAE, MSE, RMSE, and R2R^2R2.
- Plot actual versus predicted exam scores.

Task 2 — Logistic Regression

Objective: Predict whether a student passes or fails.

- Create a new target column called `Pass_Status` from `Exam_Score`.
- Set `Pass_Status = 1` if `Exam_Score >= 65`; otherwise, set it to `0`.
- Remove `Exam_Score` from the input features to prevent target leakage.
- Preprocess categorical features and scale numerical features.
- Train the Logistic Regression model.
- Evaluate accuracy, precision, recall, F1-score, and the confusion matrix.

Task 3 — K-Nearest Neighbors (KNN)

Objective: Classify students as pass or fail.

- Use the same `Pass_Status` target.
- Preprocess categorical features and standardize numerical features.
- Train KNN using k=3,5,7,9k=3,5,7,9k=3,5,7,9.
- Compare the results for each value of kkk.
- Identify the best-performing value using testing metrics.
- Explain why feature scaling is important for KNN.

Task 4 — Decision Tree

Objective: Predict whether a student passes or fails.

- Train a Decision Tree Classifier.
- Visualize the decision tree.
- Experiment with `max_depth=1, 3, 5, 10` and `None`.
- Compare training and testing accuracy.
- Identify which configurations show underfitting or overfitting.

Task 5 — K-Means Clustering

Objective: Group students according to their study habits, backgrounds, and performance factors.

- Select relevant features, such as `Hours_Studied`, `Attendance`, `Sleep_Hours`, `Previous_Scores`, and `Exam_Score`.
- Preprocess and standardize the features.
- Test K=2,3,4,5,6K=2,3,4,5,6K=2,3,4,5,6.
- Calculate the silhouette score for each value of KKK.
- Visualize the clusters using PCA to reduce the features to two dimensions.
- Describe the characteristics of each cluster.

Important: K-Means is unsupervised, so it does not require a target column. If you include `Exam_Score`, explain that the clusters are partly based on exam performance.

Task 6 — Support Vector Machine (SVM)

Objective: Classify students as pass or fail.

- Use `Pass_Status` as the target.
- Preprocess categorical features and standardize numerical features.
- Train SVM models using `linear` and `rbf` kernels.
- Compare accuracy, precision, recall, and F1-score.
- Explain the role of the kernel in separating classes.

General requirements for all algorithms

1. Load and inspect the dataset.
2. Check missing values, duplicates, and data types.
3. Perform exploratory data analysis (EDA).
4. Preprocess categorical and numerical features appropriately.
5. Split the dataset into training and testing sets.
6. Train each model and generate predictions.
7. Evaluate results using the correct metrics.
8. Compare model performance in a summary table.
9. Explain the limitations of each model and the results obtained.

Final challenge: Which algorithm performs best for predicting pass/fail status, which performs best for predicting exam scores, and which reveals useful groups of students? Support each conclusion with your experimental results.